Valerie Fong

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EDUCATION

University of California, Davis, Ph.D. in Applied Mathematics — Expected Dec 2027
Dissertation: Temporal Computation through Dynamically Critical Networks. Advisor: Rishidev Chaudhuri.

University of California, Davis, M.S. in Applied Mathematics — 2024

University of New Mexico, B.S. in Statistics and Economics, minor in Mathematics — 2022
Phi Beta Kappa

Fellowships: NSF Graduate Research Fellowship; National GEM Consortium Ph.D. Fellowship; UC Davis Hazel B. Jacoby Fellowship

RESEARCH EXPERIENCE

University of California, Davis, Graduate Student Researcher — 2022 – Present

- Developed a Python/PyTorch simulation pipeline with SLURM job arrays to study Lyapunov spectra in three recurrent network models, scaling to $N = 2{,}000$ neurons across 1,060 parameter combinations: about 25,000 simulation runs and 70,000 CPU core-hours on an HPC cluster.
- Adapted and extended a published Lyapunov-spectrum solver using the QR/Benettin method, fixed-step RK4 integration, and tangent dynamics; validated the pipeline against a reference implementation and assessed numerical convergence through paired comparisons of step size, orthonormalization interval, simulation duration, precision, and network size.
- Analyzing the resulting simulation dataset to determine how two stabilizing mechanisms shape the dependence of Lyapunov spectra on network size and gain, relative to a classical random-network baseline.
- Explored Dynamic Mode Decomposition on high-dimensional neural recordings, working with the experimental neuroscientists who collected the data to understand its biological and experimental context.

University of New Mexico, Ronald E. McNair Scholar and ASSURE Fellow — 2021 – 2022

- Applied variance-based Sobol' global sensitivity analysis to quantify how uncertainty in eight model parameters influences Hodgkin–Huxley single-neuron model outputs across hundreds of thousands of simulations.
- Presented research findings at three national McNair research conferences.

PROFESSIONAL EXPERIENCE

iRhythm Technologies, Data Science Intern — Jun – Aug 2022

- Implemented time-series forecasting approaches (ARIMA, CNNs) and decomposition-based methods (empirical mode and seasonal-trend decomposition) to analyze product demand data.
- Compared forecasting approaches across multiple error metrics and applied k-means clustering and principal component analysis (PCA) to investigate patterns in demand data.

Sandia National Laboratories, Year-Round R&D Intern — 2019 – 2021

- Consolidated and cleaned environmental, safety, and security incident data, reconciling records across multiple sources and formats.
- Built Excel visualizations and maintained a dashboard of safety, security, quality, and environmental metrics for laboratory leadership.
- Earned Lean Six Sigma Green Belt certification; held a U.S. Department of Energy Q clearance.

TEACHING

Private Tutor, Calculus, Ordinary Differential Equations, Matrix Algebra — 2026

Peer Learning Facilitator, Advanced Data Analysis I & II (University of New Mexico) — 2021 – 2022

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, PyTorch, Pandas, Matplotlib), R, MATLAB, Git, LaTeX, HPC/SLURM

Statistics and Modeling: Time-series forecasting, regression modeling, global sensitivity analysis, clustering, PCA

Numerical Methods: Numerical ODE integration, eigenspectral methods, numerical linear algebra, Lyapunov exponents, Dynamic Mode Decomposition, large-scale simulation